Chemistry · Unit 1C · Level 2 (modified)

The Strike & Staying Safe



TODAY'S PLAN

- 🔋 Energy flow
- ⚡ The strike
- 🔌 Conductors and insulators
- 🧂 Salt water
- 🏠 Safety + unit model

Review · 5 minutes

Quick Review



🧲 Like charges repel; opposites attract.

⛈️ Charge separates in a storm cloud.

❓ Unit question: Why are some places safer than others when lightning strikes?

Words to Know

Words to Know



Energy

The ability to cause change. Stored (potential) energy in the cloud turns into heat, light and sound.



Thunder

The boom made when lightning heats the air so fast that it expands.



Conductor

A material that lets electrons move through it easily, like metal or salt water.



Insulator

A material that does not let electrons move easily, like rubber, plastic or dry wood.

Words to Know

More Words to Know



🧂 **Salt water**: Salt breaks into charged particles (ions) in water, so salt water conducts electricity.

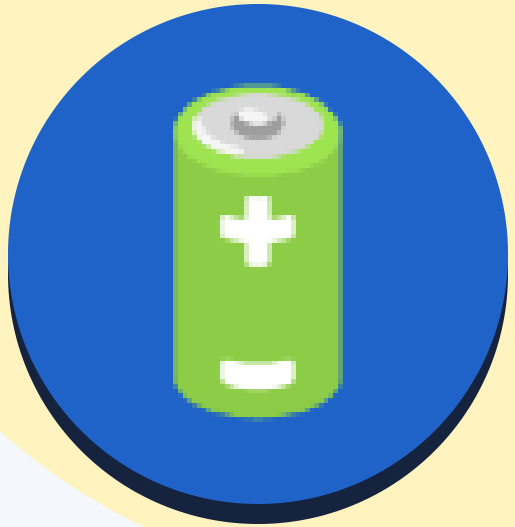
🏠 **Lightning rod**: A metal rod on a roof. It gives lightning a safe path to the ground.

🌍 **Ground**: The earth. Lightning's charge flows into the ground.

💡 **Current**: The flow of charge (electrons or ions) through a material.

Learn

Energy Flow in a Storm



➖ Separated charge stores **potential energy**, like a stretched rubber band.

⚡ When charge moves, that energy changes into **heat**, **light** and **sound**.

🌩️ Heated air expands fast: **thunder**.

🃏 Check 1 of 3



Lightning's stored energy changes into...

Hold up 1, 2, or 3:

1️⃣ Only water

2️⃣ Heat, light and sound

3️⃣ More charge

✅ Answer · Card 1 of 3



Heat, light and sound

Potential energy → thermal, light and sound.

Steps

The Strike, Step by Step



1. ➖ Negative charge builds at the cloud's **bottom**.
2. 🌩️ A **stepped leader** zig-zags down from the cloud.
3. 🌳 A **streamer** reaches up from a tall object.
4. ⚡ They connect: the bright **return stroke**.

Compare

Conductors vs. Insulators



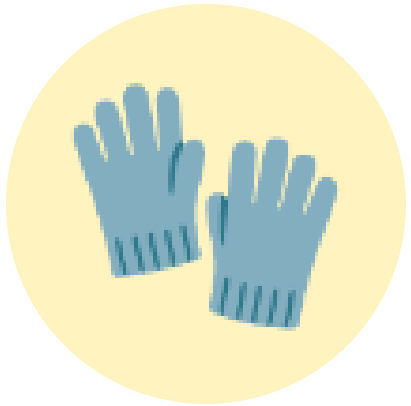
Conductors

Electrons move easily
Metals, salt water, people

Insulators

Electrons do not move easily
Rubber, plastic, glass, dry wood

🃏 Check 2 of 3



Which is an insulator?

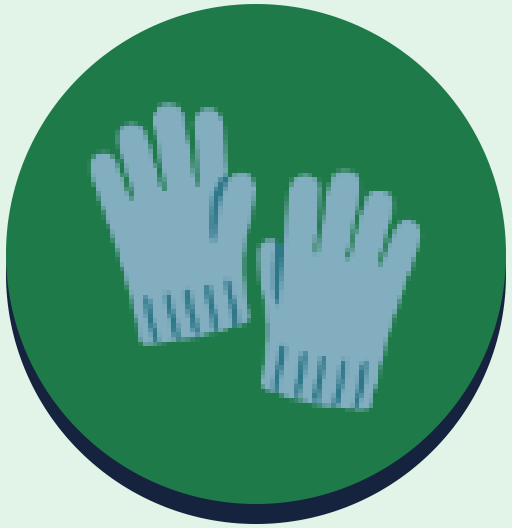
Hold up 1, 2, or 3:

1️⃣ Copper

2️⃣ Salt water

3️⃣ Rubber

✅ Answer · Card 2 of 3



Rubber

Rubber does not let electrons move easily.

Learn

No Swimming!



🧂 Salt **dissolves** in water into **ions**, charged particles.

🔌 More salt → more ions → more **current**.

🚤 Pools, lakes and the ocean all conduct. **Get out** in a storm.

🃏 Check 3 of 3



Adding more salt to water makes the current…

Hold up 1, 2, or 3:

1️⃣ Bigger

2️⃣ Smaller

3️⃣ Zero

✅ Answer · Card 3 of 3



Bigger

More ions carry more charge.

Learn

Why Some Places Are Safer

🏠 A **lightning rod** (metal conductor) carries the charge safely to the **ground**.

🚗 Buildings and hard-top cars are safe; open fields, water and tall trees are not.

📣 **When thunder roars, go indoors.**



Wrap-Up

Recap



Energy
potential → heat, light, sound



Strike
leader + return stroke



Conductor
metal, salt water



Safe place
building, car

Wrap-Up

Exit Ticket



1️⃣ What energy changes happen in a lightning strike?

2️⃣ Name one conductor and one insulator.

3️⃣ Why should you get out of a pool in a storm?